Black-Litterman Model and Robust Optimization

Risk and Uncertainty are different “unknown” measure

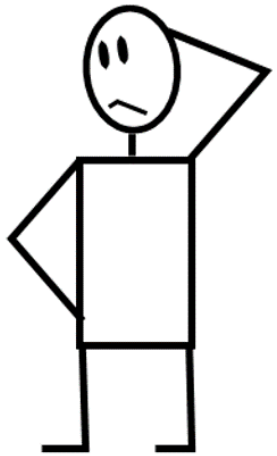
- Investors dislike both risk and uncertainty. Although both lead to uncertain outcomes, risk and uncertainty are different in nature.

- Risk is “Known unknown”
  - Must make a decision for which the outcome is not known with certainty
  - Can list all possible outcomes and assign probabilities to the outcomes

- Uncertainty is “Unknown unknown”
  - Cannot list all possible outcomes
  - Cannot assign probabilities to the outcomes

“Risk is measurable uncertainty”

“Uncertainty is unmeasurable risk”



The ideas for robust portfolio optimization

- Expected returns are difficult to forecast. The traditional portfolio optimization is usually overly reliant on expected return estimates and sensitive to small changes.

- Adding constraints can help to control the ill-effects of estimation uncertainty on optimization generated portfolio. However, this impact cannot always be mitigated by the presence of constraints.

- The uncertainty introduced by estimation error can be incorporated directly into the portfolio optimization process, referred to as robust optimization.

Motivation of Black-Litterman Model

- Traditional Markowitz Mean-Variance optimization defects:
  1. Unintuitive, highly-concentrated portfolios
  2. Input sensitivity
  3. Estimation error maximization

- Black-Litterman overcomes these problem by:
  1. Using market equilibrium expected returns
  2. Subjective views on assets, relative or absolute views both
  3. Hiring Bayesian approach to combine and form a new, mixed estimate of expected returns

Leads to intuitive portfolios with sensible portfolio weights

Market Views

Equilibrium Market

Equilibrium, is an idealized state in which supply equals demand. Although this state never actually occurs in financial markets, there're 'natural forces' in the economic system – in the form of arbitrageurs- to eliminate the deviations from the equilibrium. Hence, even though the markets are not assumed to be in static equilibrium, but it's viewed as 'centre of gravity' which can push back the deviations to it.
Equilibrium Provides a neutral reference point, leading to more reasonable and more stable optimal portfolios than portfolios based on historical mean and variance.

Why should investors take market views into consideration even if they have different views?

By saying that the investor purely proceeds with his views when he has private information, we implicitly assuming that he is **completely certain** about his private views. But that's not the case in practice.
So no matter whether there's private view about a specific asset, the investor should base his **first** insights on the return and volatility **implied by the market**.
At next step, he would update his expectation based on his **private information**.
Both of the sources are assumed to be uncertain.

Private Views

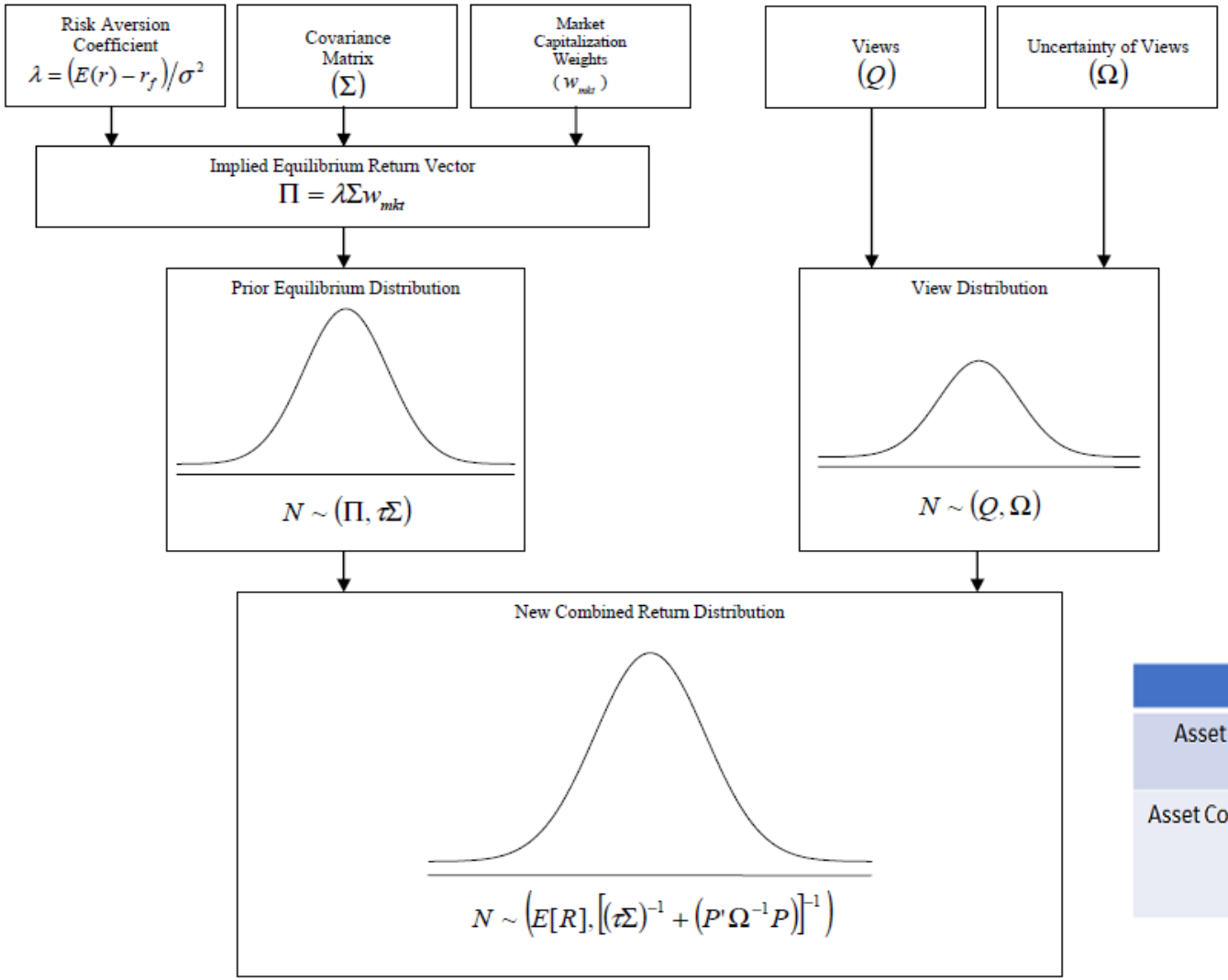
Investors can achieves higher return in semi-strong market when he as skills, which are expressed via his views.
For example, if one has a different and more precise prediction to some economic factors than the market consensus. The investor can therefore build multi-factor models to generate different forecast of assets' return and volatility form market views.
The source of the private views can be quantitative models, fundamental analysis and even a belief.

One interesting case

Assume there's a mid-sized automotive company stock XYZ and oil is relevant to XYZ stock price. Now, suppose the investor has some private information that there will be political turmoil that will affect the oil price.
Assume that the investor knows that there'll be a sanction on a significant oil supplier like Iran. The information is not publicly available. However, he estimates a 70% probability that the sanctions will be imposed.
In the case of sanctions, oil price will be raised and intuitively, a significant increasing oil price will result in lower demand for cars to great extent, lowering the profitability outlook for car producers like XYZ, in turn lower demand for its stocks would push the XYZ stock price down.

In Black-Litterman model, one specifies not only the views but also a **degree of confidence** on the views. In this example, the 70% is the confidence level. Logically, higher confidence level, smaller standard deviations and vice versa.

Black-Litterman Process



** The variance of the New Combined Return Distribution is derived in Satchell and Scowcroft (2000).*

Π is the Implied Excess Equilibrium Return Vector (N x 1 column vector);
λ is the risk aversion coefficient;
Σ is the covariance matrix of excess returns (N x N matrix); and,
w_{mkt} is the market capitalization weight (N x 1 column vector) of the assets.[4]

The risk-aversion coefficient (λ) characterizes the expected risk-return tradeoff.
τ is the scaling factor as the estimate covariance matrix is proportional to the covariance of returns Σ

	Mean-Variance Optimization	Black-Litterman Approach
Asset Mean	Mean of historical asset returns	Blended asset returns estimated from analyst views and equilibrium returns
Asset Covariance	Covariance of historical asset returns	Covariance of historical asset returns + Estimation uncertainty of the blended asset returns

Source: Idzorek, Thomas. (2005), A Step-By-Step guide to the Black-Litterman Model.

Black-Litterman Master Formula

Black-Litterman Master Formula

$$P(A|B) \sim N([(\tau\Sigma)^{-1}\Pi + P^T\Omega^{-1}Q][(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}, [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1})$$

New Expected Return (Posterior Return):

$$E[R] = \left[(\tau\Sigma)^{-1} + P^T\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P^T\Omega^{-1}Q\right]$$

Uncertainty in the posterior mean estimate (Posterior Variance):

$$M = ((\tau\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}$$

The posterior variance is the variance of the posterior mean estimate about the actual mean. It is the uncertainty in the posterior mean estimate and is not the variance of the returns.

The variance of returns:

$$\boxed{\Sigma_p = \Sigma + ((\tau\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}}$$

The estimation we feed into optimiser is $\hat{\pi}$ and Σ_p

Source: Idzorek, Thomas. (2005), A Step-By-Step guide to the Black-Litterman Model.

Black-Litterman Master Formula

$$E[R] = \left[(\tau\Sigma)^{-1} + P^T\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P^T\Omega^{-1}Q\right]$$

$E[R]$	a $N \times 1$ vector of the posterior combined return where N represents the number of assets
Π	a $N \times 1$ vector of the implied equilibrium return
τ	a scalar indicating the uncertainty of the prior estimate of the mean returns
Σ	the covariance matrix of the excess returns for the assets
P	a $K \times N$ matrix of the asset weights within each view where K expresses the number of views
Ω	a diagonal $K \times K$ matrix of the covariance of the views and means the uncertainty in each view
Q	a $K \times 1$ vector of the returns for each view

Black-Litterman Master Formula

$$E(r) = [(\tau\Sigma)^{-1} + P'\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P'\Omega^{-1}q]$$

Risk-weighted average, if P is invertible the formula is equivalent

$$E(r) = \underbrace{\boxed{\frac{(\tau\Sigma)^{-1}}{(\tau\Sigma)^{-1} + P'\Omega^{-1}P}\pi}}_{\text{Market equilibrium}} + \underbrace{\boxed{\frac{P'\Omega^{-1}P}{(\tau\Sigma)^{-1} + P'\Omega^{-1}P}P^{-1}q}}_{\text{Subjective Views}}$$

Essentially, the mean $E(r)$ is a risk-weighted average of the market equilibrium returns π and the view returns q , weighted by the inverse of the covariance matrix $(\tau\Sigma)^{-1}$ and the view uncertainty matrix $P'\Omega^{-1}$, respectively.

- The relative importance of the prior return (i.e., market equilibrium return) and the view return in the kth row of P is determined by the ratio of Ω_{kk} and $\tau p_k'\Sigma p_k$. $p_k'\Sigma p_k$ is the variance of the view portfolio, while Ω_{kk} is the variance of the kth view.
- The higher the confidence of our views (i.e., the smaller Ω), the closer our eventual return expectation $E(r)$ is to our views q.
- In the extreme example where our view is completely certain without uncertainty (or highly uncertain with a very large variance), the point estimates of E(r) are the same as our views (or market equilibrium returns)

Black-Litterman – Equilibrium View

- Black-Litterman model uses equilibrium return as the neutral starting point, that reflects current Market collective forecasts of next period returns; i.e., the market's collective view on future returns

- To do this it had ran the Markowitz mean-variance optimization and reached the optimized weights w[Nx1] – which are the current market capitalization weights

- In practice, the applicable equilibrium portfolio holding can be any optimal portfolio that the investment analyst would use in the absence of additional views on the market, such as the portfolio benchmark, an index, or even the current portfolio

Black-Litterman – Equilibrium View

- **Max $[w'\mu - (\lambda/2)w'\sum w]$**

Note: This is derived from the utility theory and multivariate normal distribution – Financial Economics 101

λ = risk aversion coefficient $(E(M) - rf)/\sigma(mkt)^2)$

E(M) = Expected market or benchmark total return

λ is found from historical data (approx = 3.07)

- Solve $\delta w'\mu / \delta w - \delta((\lambda/2)w'\sum w)/ \delta w = 0$

$\Rightarrow$ $\boldsymbol{\mu = \lambda \sum w}$

Black-Litterman – Equilibrium View

- A controversial issue in BL model.
- Since the equilibrium returns are not actually estimated, the estimation error cannot be directly derived.
- But we do know that the estimation error of the means of returns should be less than the covariance of the returns.
- A scalar τ less than 1 is used to scale down the covariance matrix (Σ) of the returns.
- Some say that "τ =0.3 is plausible".

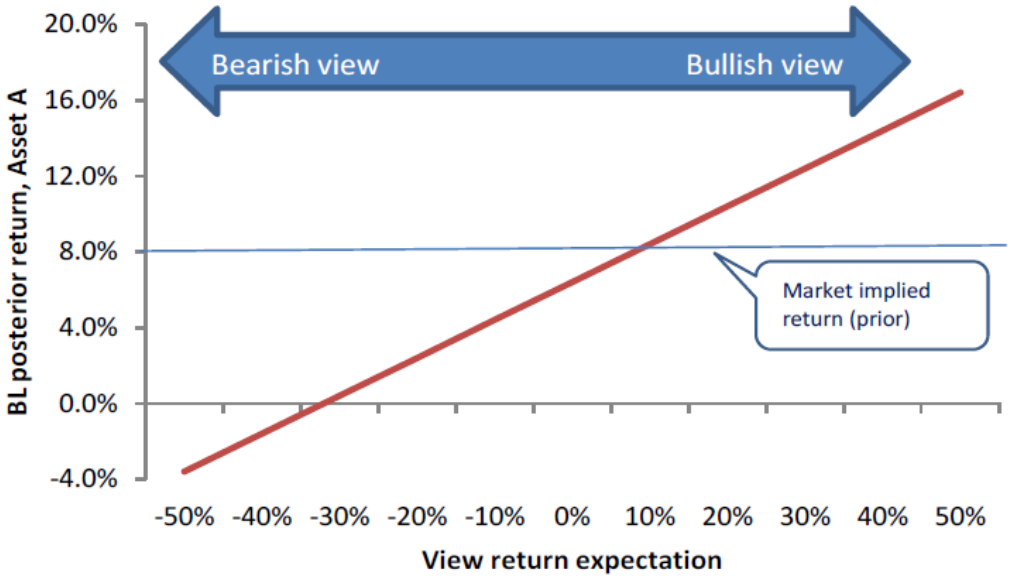
Black-Litterman – Combining views

We define the combination of the investors views as the conditional distribution.

1. By construction we will require each view to be unique and uncorrelated with the other views.
    - This will give the conditional distribution the property that the covariance matrix will be diagonal, with all off diagonal entries equal to 0.
    - We constrain the problem this way in order to improve the stability of the results and to simplify the problem.
    - Estimating the covariance between views would be even more complicated and error prone than estimating the view variances.
2. We will require views to be fully invested, either the sum of weights in a view is zero (relative view) or is one (an absolute view).
    - We do not require a view on all assets. In addition it is actually possible for the views to conflict, the mixing process will merge the views based on the confidence in the views and the confidence in the prior.

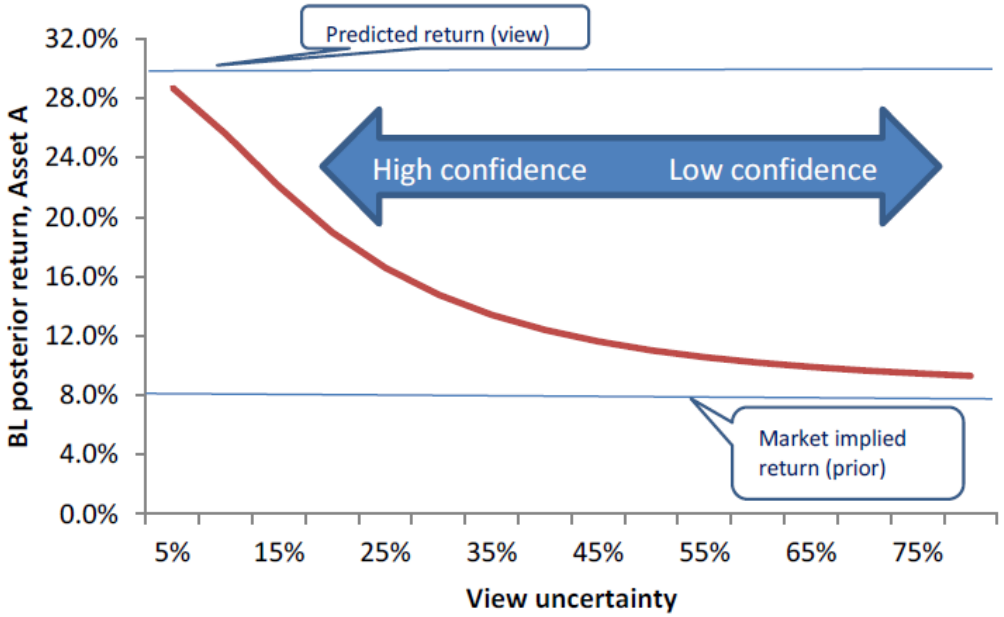
Black-Litterman – Combining views

The bullishness and bearishness of our view



Intuitively, more positive our view is, more optimistic the posterior return will be. Under some specific condition, the relationship can be linear.

Views' Uncertainty



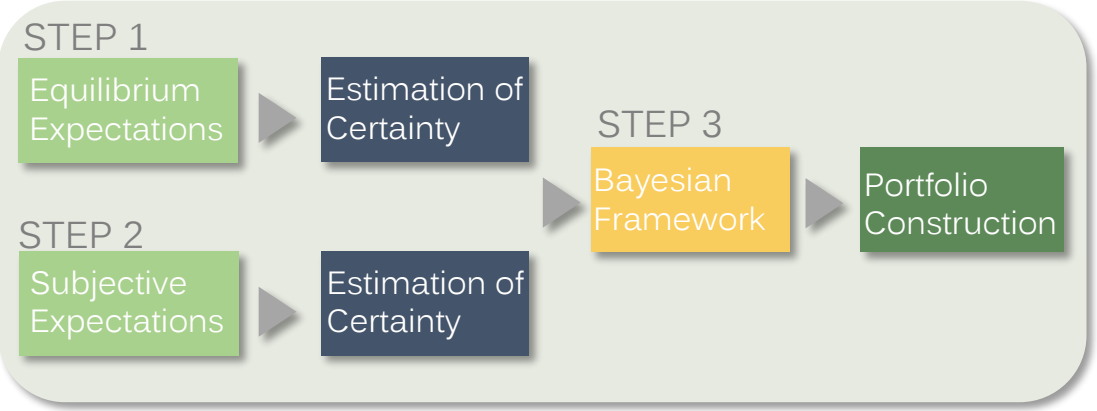
More uncertain of our views, larger the covariance matrix which will lead the posterior return more closer to market equilibrium return, vice versa.

Black-Litterman – Combining views

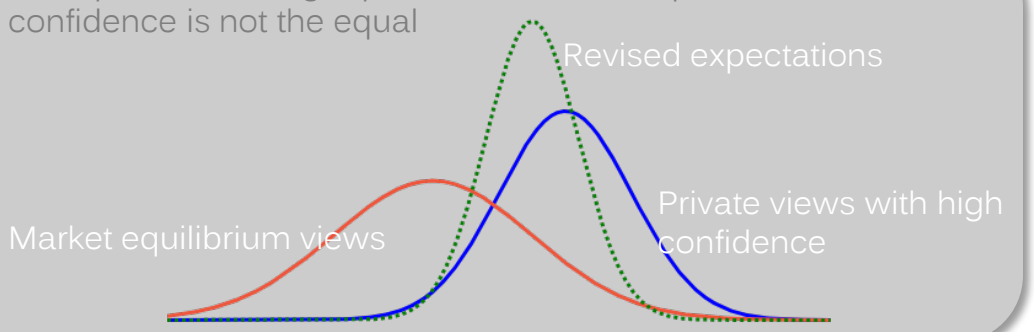
Black-Litterman model provide a method to combine the expected return implied by the **market equilibrium** with **private views**.
In a simple case, the model can be regarded as the **weighted average** of the market capitalization equilibrium portfolio and the private view portfolio. Weights are inversely proportional to the variance of the respective model. Assigning a **high variance** to a certain portfolio indicates that there's **high uncertainty** about the performance of the portfolio. Therefore, less weight should be assigned to it.

Assumptions

Assets	Investors	Markets
market and investors views on asset returns have normal distributions	are rational and behave to maximize their utility	semi-strong market efficiency hypothesis
there's risk-free assets	are risk averse and try to minimize the risk and maximum return	no transaction costs or taxes
quantities of assets are fixed and all assets are marketable and divisible	are not absolutely sure on any view	
	can estimate a level of confidence on every view	



Example of combing equilibrium views and private views where confidence is not the equal



Black-Litterman – Numerical Example

https://www.portfoliovisualizer.com/black-litterman-model

- μ_{Hist} is historical mean asset returns
- μ_p is calculated relative to the market cap. weighted portfolio using implied betas and CAPM model.
- Market portfolio weights w_{mkt} is based on market capitalization for each of the assets

Asset Class	μ_{Hist}	μ_P	w_{mkt}
US Bonds	3.15%	0.08%	19.34%
Int'l Bonds	1.75%	0.67%	26.13%
US Large Growth	-6.39%	6.41%	12.09%
US Large Value	-2.86%	4.08%	12.09%
US Small Growth	-6.75%	7.43%	1.34%
US Small Value	-0.54%	3.70%	1.34%
Int'l Dev. Equity	-6.75%	4.80%	24.18%
Int'l Emerg. Equity	-5.26%	6.60%	3.49%
Weighted Average	-1.97%	3.00%	
Standard Deviation	3.73%	2.53%	
High	3.15%	7.43%	26.13%
Low	-6.75%	0.08%	1.34%

Black-Litterman – Numerical Example

Market returns are derived from known information using Reverse Optimization:

$$\Pi = \lambda \sum w_{mkt}$$

- More return is required for more risk

$$\lambda = (E(r) - r_f)/\sigma^2 = \text{Risk Premium/Variance}$$

Using historical risk premium and variance, we got a λ of approximately **3.07**

Asset Class	μ_{Hist}	μ_P	Π
US Bonds	3.15%	0.08%	0.08%
Int'l Bonds	1.75%	0.67%	0.67%
US Large Growth	-6.39%	6.41%	6.41%
US Large Value	-2.86%	4.08%	4.08%
US Small Growth	-6.75%	7.43%	7.43%
US Small Value	-0.54%	3.70%	3.70%
Int'l Dev. Equity	-6.75%	4.80%	4.80%
Int'l Emerg. Equity	-5.26%	6.60%	6.60%
Weighted Average	-1.97%	3.00%	3.00%
Standard Deviation	3.73%	2.53%	2.53%
High	3.15%	7.43%	7.43%
Low	-6.75%	0.08%	0.08%

Black-Litterman – Numerical Example

- Opinion: International Developed Equity will be doing well
- Absolute view:
- View 1: International Developed Equity will have an absolute excess return of 5.25% (Confidence of view = 25%)
- Relative view:
- View 2: International Bonds will outperform US bonds by 25 bp (Confidence of view = 50%)
- View 3: US Large Growth and US Small Growth will outperform US Large Value and US Small Value by 2% (Confidence of View = 65%)

Black-Litterman – Numerical Example

$$\mathbf{Q} + \boldsymbol{\varepsilon} = \begin{pmatrix} 5.25\% \\ 0.25\% \\ 2.00\% \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{pmatrix}$$

- Unless a clairvoyant investor is 100% confident in the views, the error term ε is a positive or negative value other than 0
- The error term vector does not enter the Black – Litterman formula; instead, the variance of each error term (ω) does.

Black-Litterman – Numerical Example

$$
\begin{array}{cc}
 & \begin{array}{cccccccc} \text{US Bonds} & \text{Intl Bonds} & \text{US Lg Growth} & \text{US Lg Value} & \text{US Sml Growth} & \text{US Sml Value} & \text{Int'l Dev. Eqt} & \text{Int'l Emerg.Eqt} \end{array} \\
P = & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.9 & -0.9 & 0.1 & -0.1 & 0 & 0 \end{pmatrix}
\end{array}
$$

- View 1 is represented by row 1. The absolute view results in the sum of row equal to 1
- View 2 & 3 are represented by row 2 & 3. Relative views results in the sum of rows equal to 0
- The weights in view 3 are based on relative market cap. weights, with outperforming assets receiving positive weights and underperforming assets receiving negative weights

Black-Litterman – Numerical Example

$$\Omega = \begin{pmatrix} 0.0007089 & 0 & 0 \\ 0 & 0.000141 & 0 \\ 0 & 0 & 0.000866 \end{pmatrix}$$

- Ω is a diagonal covariance matrix with 0's in all of the off-diagonal positions, because the model assumes that the views are independent of each other
- This essentially makes ω the variance (uncertainty) of views

Black-Litterman – Numerical Example

$$E[R] = \left[(\tau\Sigma)^{-1} + P'\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}Q\right]$$

- First bracket "[ ]" (role of "Denominator") : Normalisation
- Second bracket "[ ]" (role of "Numerator") : Balance between returns Π (equilibrium returns) and Q (Views). Covariance $(\tau\,\Sigma)^{-1}$ and confidence $P'\,\Omega^{-1}P$ serve as weighting factors, and

$$P'\,\Omega^{-1}Q = P'\,\Omega^{-1}P\,P^{-1}\,Q$$

- Extreme case 1: no estimates ⇔ P=0: $E(R)$ = Π i.e. BL-returns = equilibrium returns.
- Extreme case 2: no estimation errors ⇔ $\Omega^{-1} \to \infty$: $E(R) = P^{-1}Q$ i.e. BL-returns = View returns.

Black-Litterman – Numerical Example

Asset Class	E[R]	Π	E[R]-Π	$\hat{w}$	$\hat{w}_{norm}$	w_{mkt}	$\hat{w} - w_{mkt}$
US Bonds	0.07%	0.08%	-0.02%	29.88%	28.83%	19.34%	10.54%
Int'l Bonds	0.50%	0.67%	-0.17%	15.59%	15.04%	26.13%	-10.54%
US Large Growth	6.50%	6.41%	0.08%	9.35%	9.02%	12.09%	-2.73%
US Large Value	4.32%	4.08%	0.24%	14.82%	14.30%	12.09%	2.73%
US Small Growth	7.59%	7.43%	0.16%	1.04%	1.00%	1.34%	-0.30%
US Small Value	3.94%	3.70%	0.23%	1.65%	1.59%	1.34%	0.30%
Int'l Dev. Equity	4.93%	4.80%	0.13%	27.81%	26.84%	24.18%	3.63%
Int'l Emerg. Equity	6.84%	6.60%	0.24%	3.49%	3.37%	3.49%	0.00%
			Sum	**103.63%**	**100%**	**100%**	**3.63%**

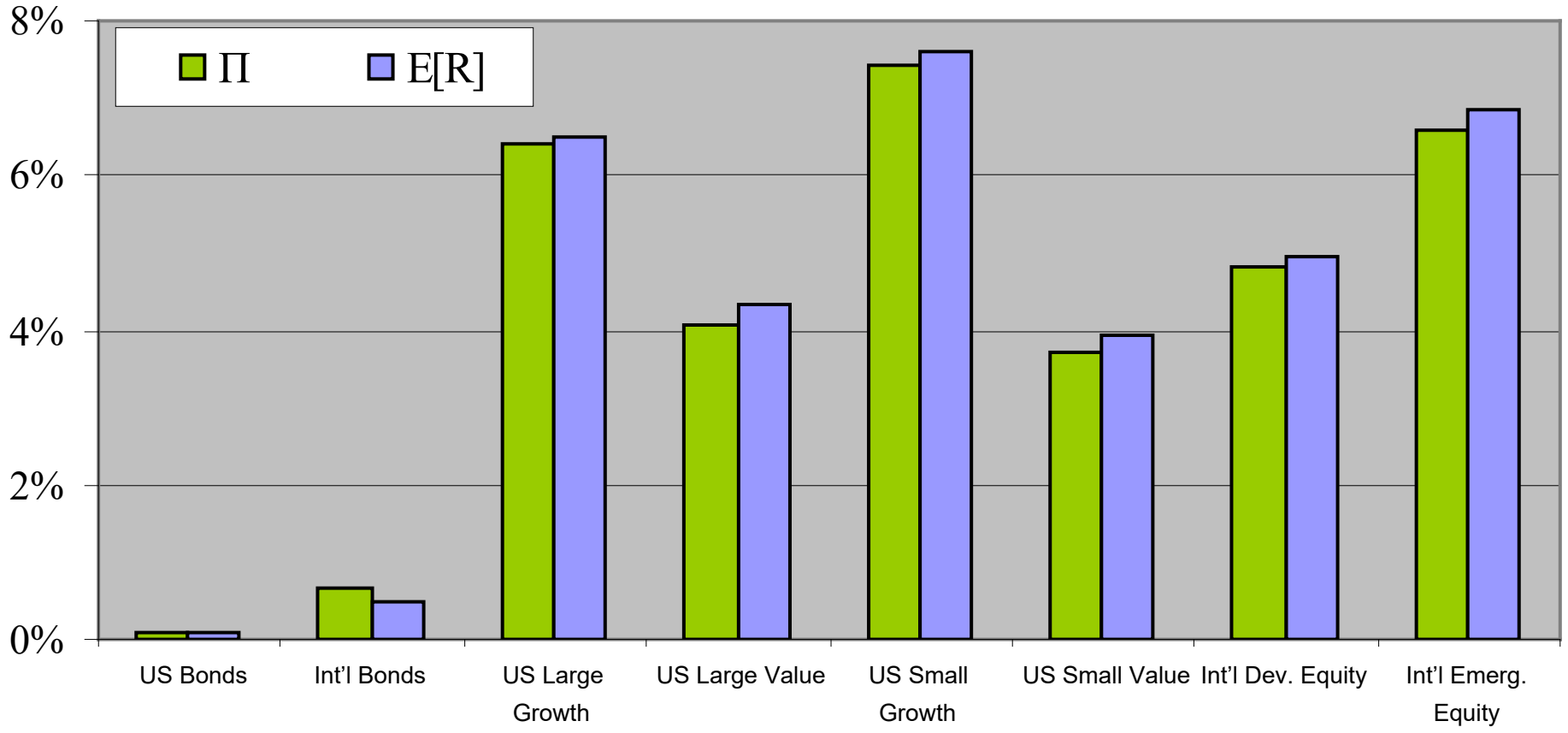
$$\Pi = \lambda \Sigma w_{mkt}$$

$$w = (\lambda\Sigma)^{-1}\Pi$$

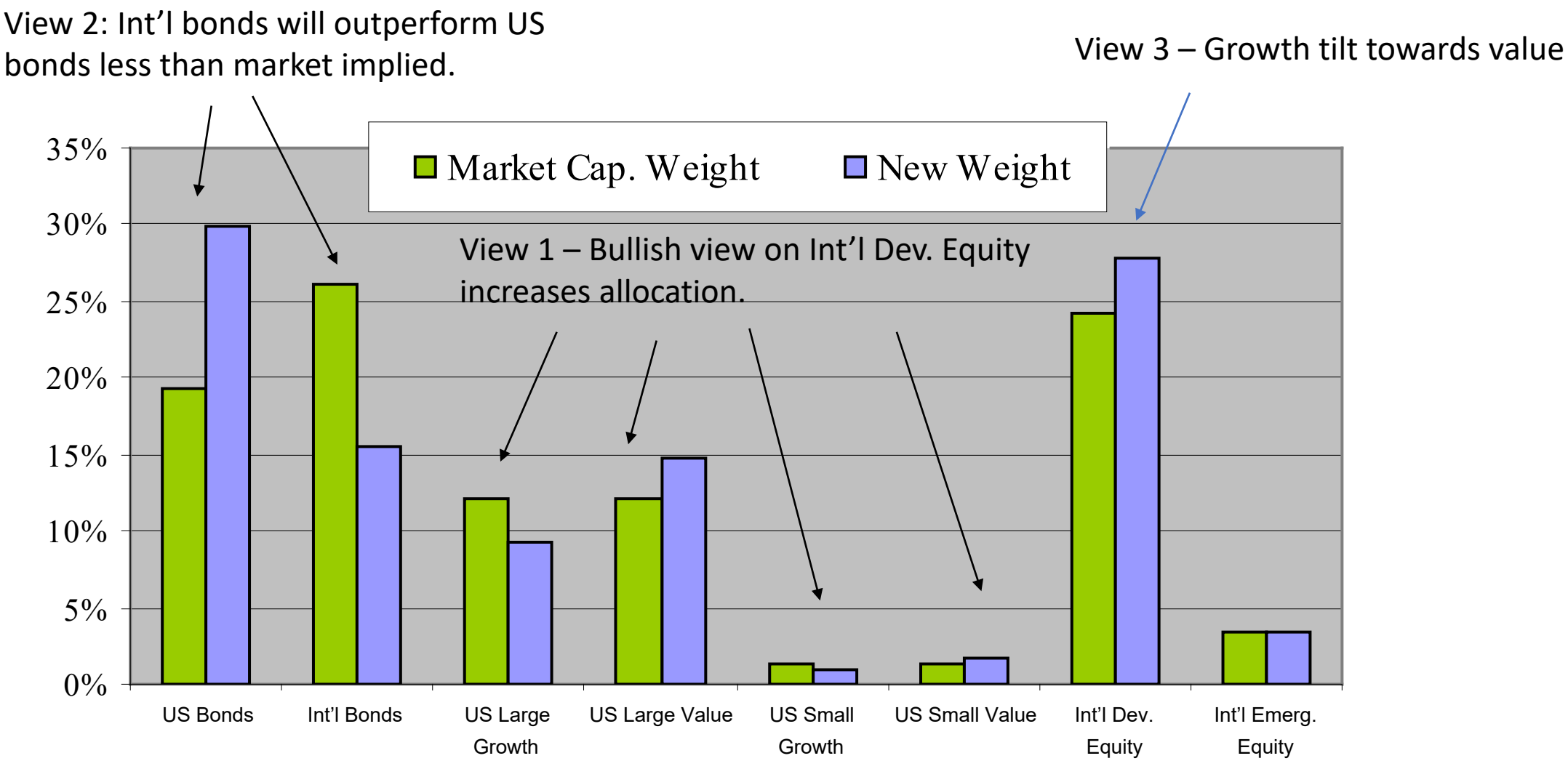
$$E[R] = \left[(\tau\Sigma)^{-1} + P'\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}Q\right]$$

$$w = (\lambda\Sigma)^{-1}E[R]$$

Black-Litterman – Numerical Example



Black-Litterman – Numerical Example

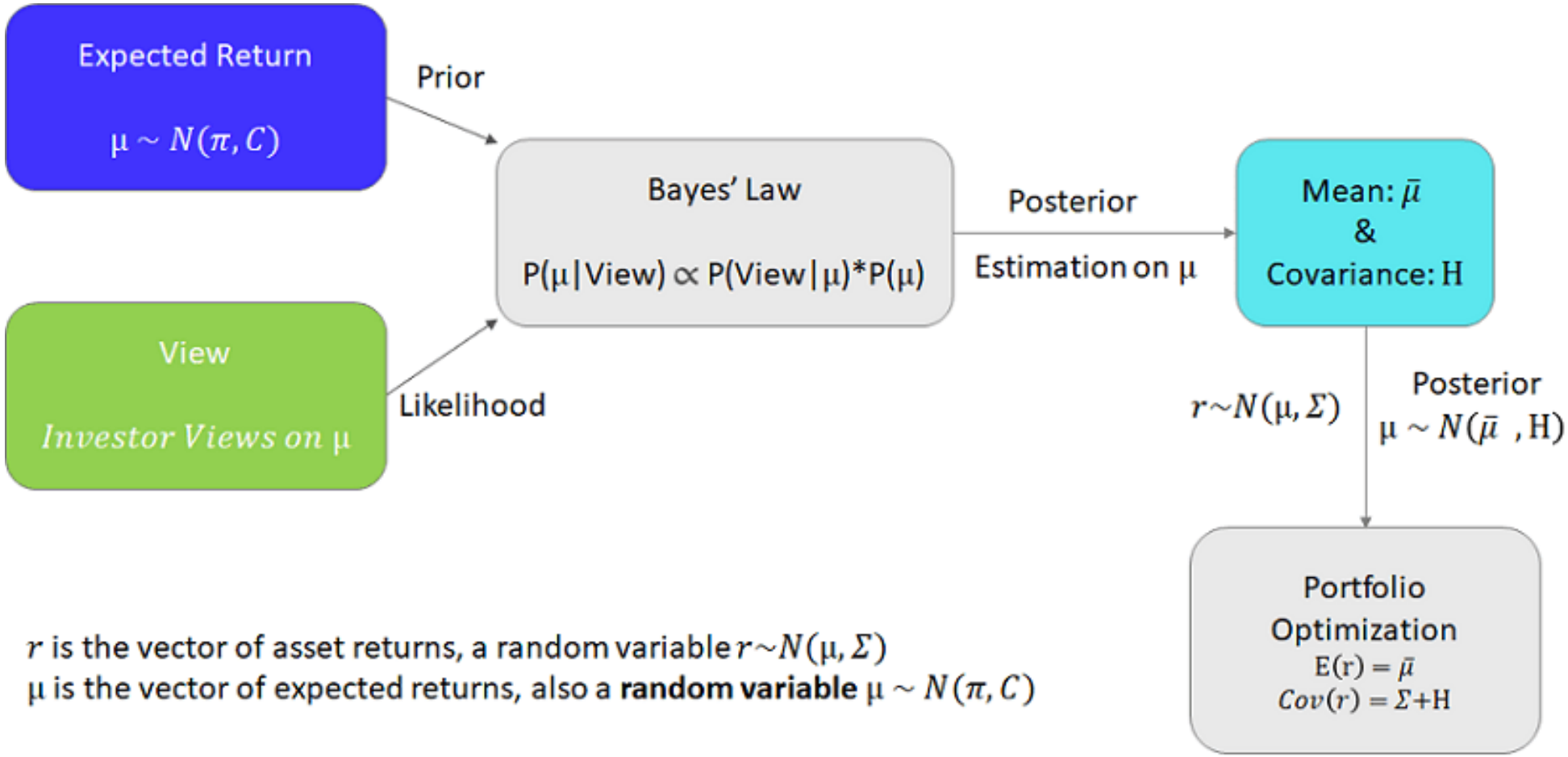


Black-Litterman Model Summary

- Advantages
  - Investor's can insert their view
  - Control over the confidence level of views
  - More intuitive interpretation, less extreme shifts in portfolio weights

- Disadvantages
  - Black-Litterman model does not give the best possible portfolio, merely the best portfolio given the views stated
  - As with any model, sensitive to assumptions
    - Model assumes that views are independent of each other

The Bayesian Definition of the Black-Litterman Model



r is the vector of asset returns, a random variable $r \sim N(\mu, \Sigma)$
μ is the vector of expected returns, also a **random variable** $\mu \sim N(\pi, C)$

Figure 1. Black-Litterman Model

The Bayesian Definition of the Black-Litterman Model

- In the context of the Black-Litterman model, posterior$\propto$likelihood$*$prior is expressed as $f(\mu|q) \propto f(q|\mu)*f(\mu)$, where each Bayesian term is defined as follows :
  - The *likelihood* is how likely it is for the views to happen given μ and is expressed as $f(q|\mu) \propto \exp[-12(P\mu-q)'\Omega-1(P\mu-q)]$.
  - The *prior* assumes the prior knowledge that $\mu \sim N(\pi,C)$ and is expressed as $f(\mu) \propto \exp[-12(\mu-\pi)'C-1(\mu-\pi)]$.
  - The *posterior* is the distribution of μ given views and is expressed as $f(\mu|q) \propto \exp[-12(P\mu-q)'\Omega-1(P\mu-q)-12(\mu-\pi)'C-1(\mu-\pi)]$.

- As previously stated, the posterior distribution of μ is also a normal distribution. By completing the squares, you can derive the posterior mean and covariance as $-\mu=[PT\Omega-1P+C-1]-1[PT\Omega-1q+C-1\pi]$, $\mathrm{cov}(\mu)=[PT\Omega-1P+C-1]-1$.

Calibrating risk aversion coefficient λ

The risk-aversion coefficient (λ) characterizes the expected risk-return trade-off. It is the rate at which an investor will forego expected return for less variance. In the reverse optimization process, the risk aversion coefficient acts as a scaling factor for the reverse optimization estimate of excess returns; the weighted reverse optimized excess returns equal the specified market risk premium. More excess return per unit of risk (a larger lambda) increases the estimated excess returns.

$$\lambda = (E(r) - r_f)/\sigma^2$$

$E(r)$ Total return of market portfolio $E(r) = w'\Pi + r_f$
r_f Risk free rate
σ^2 Variance of market portfolio $\sigma^2 = w'\Sigma w$